MECH 325 Machine Elements

Gear Train Module

Tutorial 1: Gear Trains and Kinematics

1. Adapted from **Shigley, Problem 13-16** (8th Ed., 13-14).

The double-reduction gearset shown in the figure is driven through shaft *a* at a speed of 900 rev/min. Gears 2 and 3 are spur gears and have a diametral pitch of 10 teeth/in and a pressure angle of 20°. The second pair of gears in the train, gears 4 and 5, are helical gears with a normal diametral pitch of 6 teeth/in, a 25° helix angle, and a normal pressure angle of 20° (see Figure 13-22). The tooth numbers for the four gears are: $N_2 = 14$, $N_3 = 54$, $N_4 = 16$, and $N_5 = 36$.

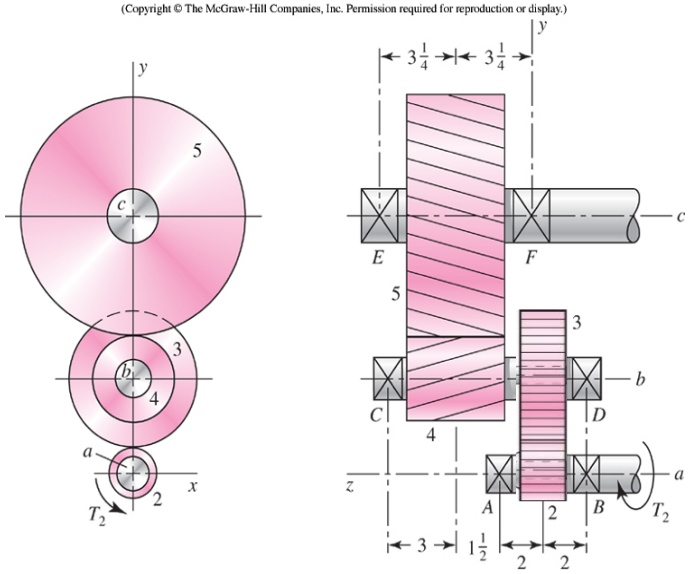
Part 1 (To be demonstrated by the TA). For the first stage (the spur gears) find:

a. The speed and direction of shaft *b*
b. The centre distance between shafts *a* and *b*

Part 2 (To be done by students). For the second stage (the helical gears) find:

c. The speed and direction of shaft *c*
d. The centre distance between shafts *b* and *c*
e. The directions of the thrust forces exerted by gears 4 and 5 on shafts *b* and *c*, respectively

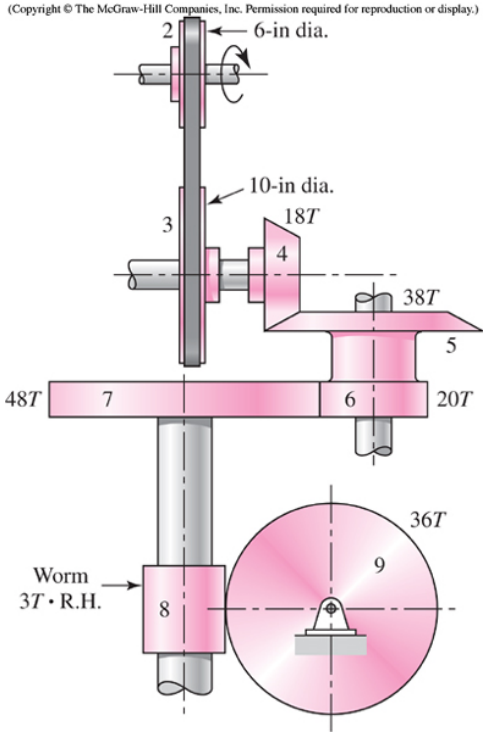
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2. **Shigley, Problem 13-18** (8th Ed., 13-16). (To be done by students.)

   The mechanism train shown consists of an assortment of gears and pulleys to drive gear 9. Pulley 2 rotates at 1200 rev/min in the direction shown. Determine the speed and direction of rotation of gear 9.

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3. Note: If time permits, continue with this problem.
   Adapted from **Shigley, Problem 13-10** (8th Ed., 13-8).

   For a spur gearset with φ = 20°, while avoiding interference, find

   Part 1 (To be demonstrated by the TA).
   - f. The smallest pinion tooth count that will run with itself (1:1 gear ratio).
   - g. The smallest pinion tooth count at a gear ratio of m_G = 2.5 and the largest gear tooth count possible with this pinion.

   Part 2 (To be done by the student).
   - h. The smallest pinion that will be run with a rack.